A Comparative Performance Study of Retrieval-Augmented Generation Systems in Gynecologic Oncology

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Abstract

Large language models (LLMs) show great potential in oncology, but their utility is limited by hallucinations and static knowledge. Retrieval-augmented generation (RAG), which grounds model outputs in curated clinical sources, can mitigate these issues. However, systematic, head-to-head evaluations of different RAG variants in oncology tasks are lacking. We compared thirteen RAG architectures with a non-RAG baseline using a single LLM (DeepSeek-R1-0528), one embedding model (BAAI/bge-m3), and a knowledge base derived from gynecologic oncology guidelines and textbooks. Two question sets were used to evaluate system performance: one focused on cervical cancer management, and the other on gynecologic oncology surgery. Each response was independently graded by gynecologic experts. Across domains, RAG systems generally outperformed the non-RAG baseline. The largest gains were observed for complex surgical questions, while improvements for guideline-based cervical cancer management questions were modest and heterogeneous. These findings demonstrate the domain-dependent value of RAG and the need for rigorous benchmarking before clinical deployment.

Introduction

Gynecologic cancers, a diverse group of cancers, represent a substantial global health burden. According to estimates from GLOBOCAN 2020, these malignancies accounted for around 1.40 million new cases and more than 670,000 deaths worldwide[1]. The management of these diseases requires a complex, multidisciplinary approach, ranging from prevention and early detection to diagnosis and highly personalized treatment strategies based on tumor histology, molecular profile, and stage[2–4]. As a consequence, the increasing complexity of gynecologic oncology creates a great information management challenge for healthcare providers.

Large Language Models (LLMs) have emerged as a revolutionary technology, rapidly permeating professional domains, including healthcare[5]. Extensive research indicates that it can enhance decision-making in clinical practice[6]. By rapidly synthesizing evidence from disparate sources, such as medical literature and clinical guidelines, LLMs can help mitigate information overload and the attendant risk of medical errors [7–9]. Moreover, these models can augment clinical reasoning: Previous studies have shown that their responses were preferred to those of physicians[10]. They are also adept at retrieving and interpreting unstructured patient data from EHRs to support enhanced personalized, evidence-informed decision-making[11].

While there is great potential for LLMs in the medical domain, large-scale use of LLMs is limited due to technical, ethical, and legal hurdles, which means that human supervision is necessary and unsupervised deployment is not allowed[12]. The primary barriers to the safe clinical integration of standalone LLMs are two inherent deficiencies. Initially, these models tend to produce false information, which is also known as “hallucination”[13]. Second, LLMs rely on static databases, which means their knowledge may be lacking in the latest medical guidelines or important information to provide correct guidance[14,15]. In a clinical setting, such deficiencies are intolerable as they can lead to diagnosis and treatment errors, which can directly threaten the safety of the patients.

The leading technology for overcoming these limitations is Retrieval-Augmented Generation (RAG). This framework helps LLMs to ground their responses on an external trusted knowledge base[16]. Its core mechanism involves first retrieving relevant information from a trusted knowledge base and then using the evidence retrieved to generate an answer. Research indicates that this method increases the accuracy and reliability of LLM outputs, reduces hallucinations, and provides transparency and citability of information used[17,18]. The RAG framework, therefore, holds immense promise for navigating the informational complexity and supporting evidence-based clinical decision-making.

The RAG system has evolved from the Naive stage into advanced frameworks with increasing sophistication[19]. Optimizations are done on various components across the pipeline. Pre-retrieval strategies such as Query Rewriting and HyDE revise the original query[20,21]. Post-retrieval methods like Rerank try to prioritize relevant documents. Self-corrective frameworks like Self-RAG and Corrective RAG (CRAG) can critique and improve outputs internally[22,23]. Despite this proliferation of techniques, there is a marked lack of systematic, standardized, head-to-head comparisons of their performance[24,25]. In addition, there are a few standardized evaluation protocols. Studies often draw on a disparate set of evaluation frameworks, each with its own limitations, and employ a mix of traditional metrics and new LLM-based evaluations[19,24]. These evaluations can be difficult to standardize due to varying settings, such as LLM versions and prompt designs. A systematic review has confirmed this gap, stating that the lack of standardized evaluation frameworks “makes it difficult to compare the performance of the RAG systems across studies,” thereby hampering the identification of the most effective RAG systems for clinical use[17].

To address this gap, this paper presents a standardized head-to-head performance evaluation of thirteen RAG variants and a non-RAG baseline in gynecologic oncology. The aim of this study is to provide evidence that guides the selection and development of RAG systems for reliable clinical decision support in specialized medical fields (Fig. 1).

Thirteen RAG variants and a non-RAG baseline were evaluated using two question sets: cervical cancer management and gynecologic oncology surgery. All questions were derived from authoritative sources, including CSCCP, ASCCP 2019, CSCO 2023, SEOM-GEICO 2023, FIGO 2018, Gynecologic Oncology Surgery (3rd ed.), and Nerve-Sparing Radical Hysterectomy. A unified experimental pipeline was applied, featuring a single curated knowledge base, BGE-M3 embeddings, standardized chunking (512 tokens, no overlap), a shared ChromaDB vector store, and the same DeepSeek-R1-0528 LLM for answer generation. Responses were independently reviewed by two experts using a four-domain rubric (relevance, evidence, accuracy, and clinical utility), and overall winrates were computed using Wilcoxon tests.

Results

Overview

Our evaluation revealed a divergence in the performance of RAG systems across the two question sets. For cervical cancer management questions, all systems, including the non-RAG baseline, achieved high efficacy with minimal performance variance. In stark contrast, when assessed on the gynecologic oncology surgery question set, a clear performance hierarchy emerged. The RAG framework proved vital in this context, with advanced architectures like Contextual Compression demonstrating a significant advantage over both the Naive RAG and the markedly inferior non-RAG baseline. The following sections provide a detailed breakdown of our findings. All generated responses are provided in Supplementary 2 Table 2 (English translations) and Supplementary 3 Table 2 (original Chinese).

Characteristics of the two question sets

We constructed two question sets and each consists of 100 items. Of these, one question set targeted cervical cancer management, and the other question set focused on gynecologic malignancy surgery. The cervical cancer question set comprised 78 questions adopted from our prior study[26] and 22 newly authored questions created from the same evidence corpus to complete the set. Surgical questions were compiled from authoritative monographs, including Gynecologic Oncology Surgery (3rd edition) and the Chinese translation of Nerve-Sparing Radical Hysterectomy[27,28]. To align with the knowledge base, all questions were originally authored in Chinese; English translations and their original Chinese versions are available in Supplementary 1 Table 4 and Supplementary 1 Table 5, respectively.

Overall evaluation framework

Two optimal chunking configurations drawing from the previous research[29] were implemented within our Naive RAG framework. All responses generated under both chunking configurations are provided in Supplementary 2 Table 1 (English translations) and Supplementary 3 Table 1 (original Chinese). The 512-token, zero-overlap configuration achieved a mean score of 2.37 (95% CI: 2.23–2.51), slightly outperforming the 256-token, 100-token overlap strategy, which scored 2.24 (95% CI: 2.08–2.40). A paired Wilcoxon signed-rank test revealed no statistically significant distinction in performance between the two chunking methods ($p = .083$). Based on this finding, the 512-token chunk size with zero overlap strategy was selected, as it offered a slightly higher mean score with greater computational simplicity. This standardized chunking configuration was subsequently applied to all RAG systems evaluated.

Performance on Cervical Cancer Management Questions

The performance on this question set appears to show a ceiling effect, as all systems evaluated show high mean scores (Fig. 2). In this scenario, the results were tightly clustered. Top-performing systems included Semantic Chunking (67.67% overall winrate; mean score 2.84; 95% CI 2.73–2.95) and Hyde (63.21%; mean 2.81; 95% CI 2.68–2.94). Notably, the Naive RAG (40.12%; mean 2.71; 95% CI 2.59–2.83) and non-RAG strategy (34.79%; mean 2.61; 95% CI 2.46–2.76) baseline also performed strongly. Although most advanced RAG techniques exceeded this strong baseline, the gap was minimal. The lowest scoring RAG method, Context Enrichment, still had a winrate of 38.64% (mean 2.64; 95% CI 2.48–2.80). This further confirms that the differences in performance were close together at the best scores.

Pairwise Wilcoxon signed-rank tests confirmed this minimal performance variance (Fig. 3). Notably, no advanced RAG system showed a statistically significant improvement over the Naive RAG baseline (all $P > .05$). Furthermore, the performance of the non-RAG strategy was statistically indistinguishable from many RAG systems, including Naive RAG ($P = .320$) and CRAG ($P = .096$). Statistically significant differences emerged primarily when comparing the top-performing system with the lowest-performing one (e.g., Semantic Chunking vs. Context Enrichment, $P = .019$) or when comparing the highest-performing RAGs with the non-RAG baseline (e.g., Semantic Chunking vs. Non-RAG strategy, $P = .010$).

The main finding in this area was the insignificant performance difference among RAG systems and non-RAG strategies. The non-RAG strategy had an overall winrate of 34.79% and the Naive RAG framework was also highly effective, achieving similar performance to the most advanced systems.

The numbers under columns A, B, C, and D represent the distribution of response quality for each grade. Error bars indicate the 95% confidence interval of the mean score, calculated using the t-distribution. Among the 14 evaluated systems, Semantic Chunking achieved the highest overall winrate of 67.67% with a mean score of 2.84 [95% CI 2.73–2.95]. Other top-tier performers included Contextual Compression (54.55%, mean 2.82), SelfRag (62.03%, mean 2.81) and Hyde (63.21%, mean 2.81). The Naive RAG baseline posted a 40.12% overall winrate (mean 2.71). Notably, the Non-RAG Strategy also performs robustly (34.79%, mean 2.61).

This heatmap illustrates the results of a pairwise comparison of performance among the 14 evaluated systems on the cervical cancer dataset. To aid interpretation, systems on both axes are ordered by their mean performance score, from highest (closest to the origin) to lowest. A Wilcoxon signed-rank test was performed for each pair of models. The color intensity in each cell corresponds to the negative logarithm of the p-value ($-\log_{10}$(p-value)), where darker red signifies a more statistically significant performance difference. The symbols denote the level of significance: insig., insignificant ($P \geq .05$); *: $P < .05$; **: $P < .01$; ***: $P < .001$, while the specific number shown is the actual P-value.

Performance on Gynecologic Oncology Surgery Questions

In contrast to the cervical cancer domain, the performance on this question set demonstrated a significant differentiation among systems (Fig. 4). The non-RAG strategy yielded the lowest overall winrate of only 20.78% (mean score 1.49; 95% CI 1.29–1.69). In comparison, the Naive RAG system alone achieved a substantially higher overall winrate of 37.43% (mean score 2.03; 95% CI 1.80–2.26).

Several advanced systems achieved even better performance, with Contextual Compression as the leading system at 84.02% (mean score 2.66; 95% CI 2.50–2.82).

Pairwise Wilcoxon signed-rank tests validated this hierarchy (Fig. 5). Contextual Compression outperformed all other systems (all $P < .05$). Furthermore, a group of high-performing systems, including SelfRAG and CRAG, significantly outperformed the Naive RAG baseline (all $P < .001$). Notably, nearly every RAG system proved statistically superior to the non-RAG baseline ($P < .001$). Nevertheless, several advanced techniques, including Semantic Chunking ($P = .763$) and Rerank ($P = .714$), did not show a statistically significant advantage over Naive RAG.

The key finding in this question set is the key value of the RAG systems. The non-RAG strategy failed (overall winrate 20.78%) while even a basic Naive RAG system provided a substantial boost. Advanced systems like Contextual Compression elevated performance further to 84.02%. This stark performance gap highlights RAG's critical role in this context.

The numbers under columns A, B, C, and D represent the distribution of response quality for each grade. Error bars indicate the 95% confidence interval of the mean score, calculated using the t-distribution. Among the 13 RAG systems evaluated, Contextual Compression achieved the highest overall winrate of 84.02% with a mean score of 2.66 [95% CI 2.50–2.82]. Other advanced RAG techniques also showed high overall winrates, including RSE (68.74%, mean 2.41), SelfRag (66.67%, mean 2.44) and CRAG (66.17%, mean 2.44). By contrast, the Naive RAG system established a baseline with an overall winrate of 37.43% (mean 2.03). While most advanced methods surpassed this benchmark, Context Enrichment was the least effective RAG strategy, registering an overall winrate of 31.61% and a mean score of 1.83 [95% CI 1.57–2.09]. Notably, the Non-RAG strategy recorded the lowest performance overall, with an overall winrate of just 20.78% and a mean score of 1.49 [95% CI 1.29–1.69].

This heatmap illustrates the results of a pairwise comparison of performance among the 14 evaluated systems. To aid interpretation, systems on both axes are ordered by their mean score, from highest (closest to the origin) to lowest. A Wilcoxon signed-rank test was performed for each pair of models, as the ordinal nature of the performance scores did not follow a normal distribution. The color intensity corresponds to the negative logarithm of the p-value ($-\log_{10}$(p-value)), where darker red signifies a more statistically significant performance difference between the two models being compared. The symbols denote the level of significance: insig., insignificant ($P \geq .05$); *: $P < .05$; **: $P < .01$; ***: $P < .001$, while the specific number shown is the actual P-value.

Comparative Analysis

Figure 6 reveals a performance divergence across the two domains. All 14 systems combined had a mean score of 2.75 (95% CI 2.72–2.79) on the cervical cancer question set. This was significantly higher than their mean score of 2.17 (95% CI 2.12–2.23; $P < .001$, Wilcoxon signed-rank test) on the gynecologic surgery question set. The disparity was even more pronounced when considering the proportion of perfect scores (A-grade rate), which was 84.64% for the cervical cancer set compared to 56.57% for the surgery set. In addition, the scores for the systems were more dispersed in gynecologic surgery (mean score 1.49–2.66, SD of means = 0.29), revealing a clear performance hierarchy where Contextual Compression (mean 2.66, 95% CI 2.50–2.82) performed best and the non-RAG baseline (mean 1.49, 95% CI 1.29–1.69) performed worst. On the other hand, in the cervical cancer domain, a ceiling effect was observed, with scores clustering near the upper limit (mean scores 2.61–2.84; SD of means = 0.07) with only minimal performance gaps among the systems.

The domain-dependent performance of the RAG architectures was obvious. Naive RAG achieved an average score that was 0.54 points higher than the non-RAG baseline, and Contextual Compression provided a further 0.63-point increase over Naive RAG. In comparison, the differences in the cervical cancer domain were slight: Naive RAG improved the average score over the non-RAG baseline by only 0.10 points, and the best-performing system, Semantic Chunking, yielded just an additional 0.13-point gain over Naive RAG. This is corroborated by our statistical findings: in the surgery domain, advanced systems like Contextual Compression, SelfRAG, and CRAG significantly outperformed Naive RAG ($P < .001$), whereas no such advantage was observed in the cervical cancer domain (all $P > .05$).

We also conducted two-sided Mann-Whitney U tests between the score distributions of all evaluated systems on the two question sets to measure the magnitude of cross-domain performance shifts (Fig. 7). The diagonal cells, which compare each system's performance across the two question sets, show that most systems exhibited a noticeable shift, for instance, the performance of Non-RAG, Naive RAG, Semantic Chunking, and HyDE considerably differed ($P < 001$). In contrast, the performance of Contextual Compression did not differ significantly. The Off-diagonal cells report cross-architecture and cross-domain contrasts, confirming the performance disparity between the two question sets.

To sum up, these results show (i) tight clustering and marginal RAG gains for cervical cancer management, and (ii) broad dispersion with a clear hierarchy and obvious RAG advantages in gynecologic surgery.

Side-by-side diverging bar charts summarize overall winrate (bar length; fixed gridlines at 0/20/40/60/80%), with methods ordered within each panel by descending overall winrate. Grades A/B/C/D were mapped to 3/2/1/0; means and 95% CIs are shown at the margins. A clear cross-set contrast emerges: the Gynecologic Surgery set is harder overall (Non-RAG strategy mean 2.61→1.49; Naive RAG 2.71→2.03). Advanced architectures diverge in transferability—Contextual Compression improves markedly (54.55%→84.02%; mean 2.82→2.66), and RSE, CRAG, and Self-RAG remain among the top performers—whereas Semantic Chunking and HyDE drop (67.67%→39.17%, 63.21%→42.88%). Accordingly, the cervical set shows a compressed distribution (winrates ranging from 35% to 68%; means 2.61–2.84), while the surgery set is widely dispersed (20.78%–84.02%; 1.49–2.66), with Contextual Compression highest and the non-RAG baseline lowest.

Each cell shows the two-sided Wilcoxon rank-sum p-value comparing per-question score distributions between gynecologic oncology surgery (rows) and cervical cancer management (columns). Diagonal cells compare the same system across domains; off-diagonal cells compare different systems across domains. Color encodes $-\log_{10}(p)$. Annotations in each cell display the mean score for the row (Surgery) versus the column (Cervical), followed by significance markers (insig., $P \geq .05$; *: $P < .05$; **: $P < .01$; ***: $P < .001$).

Discussion

To remedy the absence of standardized comparisons among RAG architectures, this study presents a head-to-head evaluation of thirteen RAG variants. Our observations show that the value and optimal architecture of RAG systems are task dependent.

The task-dependent value of RAG is evident from the performance difference between our two question sets. In the highly specialized domain of gynecologic oncology surgery, the non-RAG strategy, which relied solely on parametric knowledge, performed poorly. However, even a Naive RAG implementation provided a substantial performance boost. Advanced architectures, such as Contextual Compression, offered a further uplift, showing a statistically significant advantage over Naive RAG systems.

However, RAG's advantage diminished in the cervical cancer management questions. The non-RAG strategy performed exceptionally well and all RAG systems—Naive or advanced—yielded only marginal gains. Crucially, no advanced RAG system yielded a statistically significant improvement over Naive RAG.

According to these findings, RAG systems should be specifically selected based on the needs of the clinical task. This means using high-performing architectures such as Contextual Compression for highly specialized domains, while using simpler architectures for more common and guideline-concordant questions. This strategy helps boost performance and reduce the complexity and cost of RAG systems, where the improvement is proven to be minor.

The challenging surgical domain revealed a distinct performance hierarchy among RAG architectures. Naive RAG provided a robust baseline; contrary to expectations, many advanced RAG architectures, such as Reranking and Semantic Chunking, failed to yield statistically significant improvements. While other architectures, especially the Contextual Compression, showed significant improvements. This finding challenges the conventional assumption that advanced techniques are superior to simpler ones in terms of performance[30]. It indicates that for high-stakes, specialized applications, simply adopting an advanced RAG is insufficient; a careful selection of an architecture proven to be effective for the target knowledge type is vital to achieving expert-level accuracy.

These performance patterns can be better understood by examining the interplay between the model’s parametric knowledge and the structure of the external knowledge sources. The high accuracy of the non-RAG strategy on the cervical cancer questions indicates LLM’s robust internal knowledge on common medical topics, a finding consistent with previous studies[10,31–33]. However, it performed much worse in the more specialized domain of gynecologic oncology surgery. This disparity is consistent with previous research[34,35], which demonstrated that an LLM's recall is influenced by a topic's prevalence in its training data. This is likely rooted in the pre-training process, which largely relies on vast corpora of publicly accessible internet text. For instance, the training data for the landmark model GPT-3 was largely composed of web-crawled data (e.g., Common Crawl) and sources like Wikipedia, which in total accounted for over 85% of its training corpus[36]. Clinical guidelines for cervical cancer are well-accessible online, while the knowledge for surgery is usually found in copyrighted monographs that are not as accessible. The accessibility bias in the training data explains the model’s robust knowledge for the cervical cancer management questions and sparse f knowledge of gynecologic surgical oncology.

The task-dependent value of RAG systems can also be attributed to this disparity. For the cervical cancer questions, where the LLM's parametric knowledge is already strong, even advanced RAG architectures yield only marginal gains. Conversely, for the knowledge-sparse surgical domain, RAG systems bridge the gaps in the model's parametric knowledge. This is further supported by research[37] from Basmov et al., which demonstrates that an LLM's ability to comprehend contexts deteriorated when processing 'imaginary' data—fictitious entities which were not included in, yet not contradictory to the model's parametric knowledge. Their work further showed that this deterioration happened especially when the LLMs were processing semantically complex texts. In our experiment, the gynecologic oncology surgery questions are rich in procedural and conditional logic, which is highly semantically complex.

In addition, the performance gaps among RAG systems also stem from the differences in the knowledge structures of the two domains. The cervical cancer knowledge base derived from clinical guidelines, which is structured, declarative, and rule-based (e.g., "if condition X, then action Y")[38–42]. Such organized information has been proven to enhance LLM reasoning and answer accuracy[43]. Consequently, even Naive RAG can provide sufficient, high-quality contexts, leading to a performance ceiling. In stark contrast, the gynecologic oncology surgery questions demand procedural and visuospatial knowledge[27,28]. The descriptive, non-algorithmic nature of these contents results in dense, narrative-style retrieved contexts. This creates a high noise-to-signal ratio where critical facts are interwoven with explanatory text. In this challenging context, the ability of an RAG system to filter and interpret unstructured information becomes crucial, explaining the significant performance gaps observed among systems in the surgical domain.

Our findings concur with existing research highlighting the importance of RAG in improving LLM performance and mitigating model-generated hallucinations in medical contexts[44,45].The evaluation results for top-performing systems like Self-RAG, CRAG, and Contextual Compression align with their reported strengths in other studies.[22,23,46] The superior performance of contextual compression may be attributed to its ability to filter and condense retrieved information, which helps prevent the "lost-in-the-middle" problem where LLMs struggle to utilize information embedded deep within a long context[46,47]. By extracting the most relevant sentences, it reduces the high noise-to-signal ratio in the narrative surgical texts and presents the LLM with a cleaner, more focused context from which to derive complex procedural steps. This precision is less critical for the structured, rule-based cervical cancer knowledge but becomes critical in the surgical domain.

Similarly, the success of CRAG and Self-RAG in the challenging surgical domain can be attributed to their shared principle of adaptive, self-revision retrieval. The CRAG framework first enables the language model to determine retrieval necessity, then evaluates the relevance of initial documents, and dynamically augments the context with web search results when retrieved information is deemed insufficient[23]. Self-RAG incorporates a dedicated evaluator to assess the quality of the retriever's output and triggers distinct corrective actions[22]. The surgical knowledge is complex and its textual representation can be ambiguous, making naive retrieval prone to providing context that is only superficially relevant. The evaluation components within CRAG and Self-RAG act as a critical quality control gate. They assess whether the retrieved documents are genuinely useful for answering a specific, nuanced question, rather than just being topically related. This self-correction is proven indispensable within the complicated surgical domain. The advanced architectures of these systems, which add a layer of metacognitive evaluation to the retrieval process, are the primary drivers of their enhanced performance.

On the other hand, our study also yielded several unexpected findings. Advanced RAG techniques, such as Hyde, Reranking, and Query transformation, did not significantly outperform Naive RAG in our study. But in the literature,[48–50], they have been shown to significantly enhance RAG system performance. Moreover, the Context Enrichment architecture consistently failed. Across both question sets, it ranked among the least effective methods.

The subpar performance of several advanced techniques can be attributed to specific architectural limitations. For Reranking, which mainly optimizes retrieving relevance, our results may reflect a known trade-off where optimizing solely for retrieval relevance can paradoxically degrade the quality of the final generated answer[51]. Similarly, Semantic Chunking showed no significant improvement, suggesting that its benefits are highly context-dependent. Its effectiveness is proven to be most pronounced on artificially constructed documents with high topic diversity, whereas it can underperform simpler methods on more realistic, topically coherent texts[52]. The Context Enrichment, which shows the worst average score, suggests that merely adding more adjacent context can introduce “noise” and degrade performance by flooding the context window[53]. The negative impact of this “noise” is more precisely defined as the “distracting effect” by Amiraz et al[54]. According to their study, the adjacent chunks can act as 'hard distracting passages'—semantically related yet irrelevant information that actively misleads the LLMs. Their research further confirms that introducing such passages, even alongside the correct answer, causes a drop in accuracy.

Pre-retrieval strategies also failed due to a mismatch with our study's precise, professional questions. Automatic query expansion (AQE), the underlying methodology of Query Transformation, was designed for short, ambiguous user queries[55]. This design principle prioritizes breadth over precision, which lowers total relevance scores even on the general queries it was built to improve[56]. This results in contextual dilution: the highly relevant documents retrieved by the original query are mixed with a larger volume of tangential information, which impairs LLM’s ability to generate an accurate response.

Likewise, HyDE's effectiveness, though proven in other contexts [57], depends on the LLM's ability to generate a factually correct hypothetical document. In the knowledge-sparse surgical domain, this premise failed: the model does not have sufficient knowledge to produce a representative samples[58], which in turn misled the retrieval process by searching for content aligned with its own misconceptions rather than the user's intent.

In contrast to the inconsistent gains from these advanced techniques, our findings show that the Naive RAG framework is surprisingly robust. While it was surpassed in the complex surgical domain, its implementation marked a monumental leap in performance over the non-RAG strategy, boosting the mean score from 1.49 to 2.03. Moreover, in the more common-knowledge domain of cervical cancer, Naive RAG also performed exceptionally well.

Beyond these mechanistic explanations, our findings support a task-dependent, cost-benefit framework for deploying RAG systems in clinical environments. For high-frequency, low-risk tasks, such as answering guideline-related questions where the LLM's parametric knowledge is often sufficient, the Naive RAG system offers a computationally efficient solution. Even if our study showed that it did not significantly beat the non-RAG baseline, its low deployment overhead justifies its use as an essential "guardrail" against potential LLM hallucinations. On the other hand, advanced RAG architectures are required for high-stakes specialized uses such as surgical planning. In these situations, the substantial performance improvements achieved by systems such as Contextual Compression are critical for clinical safety and utility, thereby justifying their greater computational cost and latency.

For clinical utility, RAG systems must function as "cognitive co-pilots," augmenting rather than replacing clinician judgment within a human-in-the-loop framework. Practical applications include embedding RAG into an Electronic Health Record (EHR) for guideline queries or using it to support pre-operative surgical planning. This partnership mandates a clear ethical framework, foremost addressing accountability. The final clinical decision-making responsibility must reside with the physician; the RAG system's role is to provide informational synthesis, not medical directives. While RAG's inherent transparency—its ability to "show its work" by citing sources—is a key ethical advantage for responsible use, institutions must also guard against reproducing biases present in the underlying literature. Thus, successful deployment requires pairing technical integration with an ethical framework that preserves human oversight and accountability.

Our findings, particularly the unexpected underperformance of certain advanced RAGs, should serve as a cautionary tale against adopting architectures based solely on general-purpose benchmarks, as doing so could compromise patient safety. We propose that our method offers a replicable validation template: institutions should (1) specify the clinical use case before deployment, (2) construct a knowledge base of trusted local sources, (3) create a representative question set with clinical experts, and (4) conduct a head-to-head evaluation to identify the most suitable system for that specific task.

Furthermore, this validation is not a one-time event. A RAG system's reliability depends on its knowledge base, which must be treated as a "living" asset. In other words, we have to use a lifecycle approach that combines pre-deployment validation with post-deployment continuous monitoring. So, it means frequent updates of source materials, a mechanism for acquiring user feedback and recurrent validation using a gold-standard question set to check for “performance drift” due to knowledge base or underlying LLM updates. Ongoing governance is necessary to ensure the safety and effectiveness of utilizing RAG systems in clinical practice.

Several limitations of this study should be acknowledged. Our knowledge base was intentionally restricted to five reputable Chinese-language sources. Consequently, our findings may not be generalizable to scenarios involving larger, more heterogeneous, and multilingual knowledge bases. Moreover, the text-only nature of the corpus limits its ability to represent procedurally significant visuospatial knowledge, a critical challenge in surgical domains.

The generalizability of our quantitative results is limited by the use of a single LLM (DeepSeek-R1-0528) and embedding model (BAAI/bge-m3). The observed performance ranking is contingent on the parametric knowledge of the LLM employed. Consequently, a different model with a stronger or weaker baseline knowledge in gynecologic oncology could alter the performance gap between RAG and non-RAG strategies, as well as the relative rankings of the various RAG architectures.

Our assessment framework is strong, but not perfect. The 4-point ordinal scoring system was reliable, but it may have produced the ceiling effect observed in the cervical cancer domain, potentially masking subtle performance differences. Besides, pre-defined questions do not represent the iterative, ambiguous, and contextual character of genuine clinical questions accurately.

Despite these limitations, our core conclusion remains robust. The study's primary contribution is the establishment of a key design principle for clinical AI: the necessity and complexity of an RAG system should be tailored to the knowledge specificity of the clinical domain. This principle serves as a robust and practical heuristic for developing more reliable AI applications in medicine.

Finally, several directions for future research emerge from our findings. Future work needs to further validate our findings. It is essential to test a broader range of LLMs and embedding models. Also, extending the knowledge base to larger multilingual multi-source (e.g., journal articles, clinical trials) corpora is also necessary to evaluate performance in a more realistic and noisier information setting.

We must move from evaluations in simulation circumstances to evaluations in the clinical setting. This includes building prototype clinical decision support systems and conducting human-in-the-loop usability studies with clinicians. Future randomized controlled trials are also needed to evaluate the effects of these systems on clinical decision-making accuracy, workflow efficiency, and patient outcomes.

Our findings highlight the limitations of text-only RAG technologies for procedural domains like surgery. Therefore, future research should focus on developing multi-modal RAG systems. Such systems would retrieve and synthesize information from a diverse knowledge base containing not only text but also anatomical diagrams, intraoperative images, and surgical videos. By integrating these modalities, they could provide context-rich, actionable answers to complex procedural questions, potentially transforming surgical training and preoperative planning.

To fill the gap in the comparative literature on RAG architectures, this work evaluates thirteen architectures in the context of gynecologic oncology. We find that the benefit of RAG is extremely task-dependent: it is critical in specialized, knowledge-sparse domains, such as surgery questions, but offers little advantage in common-knowledge domains, such as cervical cancer management questions. In the surgery domain, the performance hierarchy was clear and it tells a key message that the complexity of an RAG architecture must match the clinical application's knowledge specificity. Future research should confirm this finding in other models and knowledge bases, translate them to real-world clinical trials, and develop a multi-modal RAG to overcome the limitations of text-only systems in complex procedural tasks.

Methods

Study Design

In this study, thirteen RAG systems and a non-RAG baseline were evaluated. We fixed the backbone LLM, embedding model, and the knowledge base to isolate the impact of RAG architecture on answer quality. Two complementary question sets were developed: One focused on cervical cancer management and the other on the specialized gynecologic oncologic surgery. Response quality was graded with a four-level rubric by expert gynecologists, and statistical analysis was then conducted. Our methodology is detailed in the following sections.

Question Development

All questions stemmed from authoritative clinical sources to ensure guideline concordance. The framework (principles, structure, and answer-keying strategy) of question design was adopted from our previous publication and uniformly applied across both question sets[26]. The questions emphasize guideline-concordant, decision-focused content suitable for objective scoring.

Cervical Cancer Questions

This question set focused on the screening, diagnosis, and treatment of cervical cancer. Specifically, cervical cancer screening content primarily drew on the Chinese Society for Colposcopy and Cervical Pathology (CSCCP) Consensus on cervical cancer screening and abnormal management in China[38], the 2019 American Society for Colposcopy and Cervical Pathology (ASCCP) Risk-Based Management Consensus Guidelines for Abnormal Cervical Cancer Screening Tests and Cancer Precursors [39], and supplemented by the Chinese Society of Clinical Oncology (CSCO) Guidelines for the Diagnosis and Treatment of Cervical Cancer (2023)[40].

Cervical cancer diagnosis and treatment content referenced the Sociedad Española de Oncología Médica-Grupo Español de Investigación en Cáncer de Ovario (SEOM-GEICO) Clinical Guidelines on Cervical Cancer (2023)[41], the CSCO guideline [40], and the International Federation of Gynecology and Obstetrics (FIGO) 2018 Gynecologic Cancer Report – Interpretation of the Cervical Cancer Guidelines[42].

Gynecologic Oncology Surgery Questions

This question set focused on surgical procedures in gynecologic oncology. These questions mainly drew on Gynecologic Oncology Surgery (3rd edition)[27] and the Chinese translation of Yabuki's monograph Nerve-Sparing Radical Hysterectomy[28].

Knowledge Base Construction

The knowledge base was assembled from five primary sources: Nerve-Sparing Radical Hysterectomy [28], Gynecologic Oncology Surgery (3rd edition)[27], the Chinese Society of Clinical Oncology (CSCO) Cervical Cancer Guideline 2023[40], the CSCCP expert consensus interpretation on cervical cancer screening and management of abnormal results[38], and the FIGO 2018 Gynecologic Cancer Report—Cervical cancer guideline interpretation[42]. Their PDF editions were first converted to UTF-8 texts using the open-source opendatalab/MinerU[59] and then all texts underwent manual proofreading to correct parsing artifacts. The cleaned texts were stored, forming a single canonical repository used across all RAG systems.

Model Configuration

Large Language Model

We used the open-source DeepSeek-R1-0528 as the answer-generation model for all RAG variants. DeepSeek-R1-0528 is a minor version upgrade of DeepSeek-R1, which enhances the depth of reasoning and inference. According to the official report, its overall performance approaches contemporary frontier LLMs. The model shows competitive performance on reasoning-intensive benchmarks (mathematics, programming, general logic) and substantial gains on AIME 2025 (accuracy increase from 70% to 87.5%) accompanied by longer average reasoning traces (≈ 23k vs 12k tokens), with additional reductions in hallucination. In our experiments, the model was accessed via the official DeepSeek cloud API, and identical API parameters were used across all RAG conditions.

Text Embedding and Chunking

All RAG systems used a unified embedding and text chunking process. For vectorization, we employed the BAAI/bge-m3 model, a versatile embedding model that supports multi-functional, multi-lingual, and multi-granularity retrieval. Moreover, this model allows text input size up to 8192 tokens, which makes it suitable for our diverse medical corpus.

A preliminary validation study was conducted to determine the chunking strategy. Based on previous research[29], we compared two top-performing configurations: (1) a chunk size of 1024 tokens with zero overlap (using TokenTextSplitter); (2) a chunk size of 512 tokens with a 200-token overlap (using SentenceSplitter). For both strategies, token counts were calculated using the cl100k_base tokenizer. These settings, when applied to our knowledge base, correspond to an approximate chunk size of 512 tokens with zero overlap and 256 tokens with a 100-token overlap, respectively, as processed by the tokenizer native to the Deepseek-R1-0528 model.

We implemented both configurations within our Naive RAG pipeline and evaluated their performance on our two question sets. The generated answers were assessed using the four-level scoring rubric (A-D) as described in the "Scoring Process and Criteria" section. We then used a paired Wilcoxon signed-rank test to determine if there was a statistically significant performance difference between the two chunking configurations.

RAG System Architectures and LLM Baseline

We sourced candidate RAG pipelines from a high-star, community-adopted open-source repository[60]. From this repository, we selected 13 distinct and representative RAG variants chosen to cover a comprehensive range of architectural enhancements across the RAG pipeline. Specifically, our selection encompassed: two Pre-retrieval Enhancement Strategies (HyDE and Query Transformation RAG); three Indexing and Chunking Strategies (Semantic Chunking, Context Chunk Headers, and Document Augmentation RAG); two Core Retrieval Strategies (Recursive Small-to-big Embedding and Fusion RAG); three Post-retrieval Refinement Strategies (Context Compression, Rerank, and Context Enrichment RAG); and two Self-Corrective Frameworks (Corrective RAG and SelfRAG). The overview of these systems is provided in Table 1. All 13 systems used ChromaDB as the uniform vector database for storing document embeddings.

We also evaluated the DeepSeek-R1-0528 model in a non-RAG configuration. In this condition, the model was used to answer the same question sets directly, without access to any external knowledge base. This allows us to measure the performance of the LLM's parametric memory. All detailed workflows, prompts, and parameters for every system are provided in Supplementary 1 Fig. 1–13 and Table 1–3.

Table 1
The Description Of Included RAG Systems

System	Category	System Description	Architecture
Naive RAG	Baseline System	This RAG system splits texts into fixed-size chunks, embeds each chunk into a vector representation, and retrieves the top-k candidates based on vector similarity to a given query.	Supplementay1 Fig. 8
HyDE	Pre-retrieval Enhancement Strategies	This RAG system embeds an LLM-generated hypothetical answer for each query and then uses this embedding to retrieve relevant chunks via vector similarity.	Supplementay1 Fig. 7
Query Transformation RAG	Pre-retrieval Enhancement Strategies	This RAG system enhances retrieval by reformulating the initial user query into multiple variants—including rewritten, step-back, and decomposed sub-queries—and then aggregating the search results from all variants to construct a more comprehensive context for the language model.	Supplementay1 Fig. 9
Semantic Chunking RAG	Indexing and Chunking Strategies	This RAG system segments texts by identifying natural topic shifts, creating boundaries at points where the measured semantic similarity between adjacent sentences drops below a statistically defined threshold.	Supplementay1 Fig. 13
Context Chunk Headers RAG	Indexing and Chunking Strategies	This RAG system enhances retrieval by enriching chunks with LLM-generated titles, which are prepended before embedding to improve the contextual accuracy of both the retrieval and generation stages.	Supplementay1 Fig. 2
Document Augmentation RAG	Indexing and Chunking Strategies	This RAG system generates multiple synthetic questions for each document chunk and co-embedding both the original text and the questions into the vector store.	Supplementay1 Fig. 5
RSE (Recursive Small-to-big Embedding)	Core Retrieval Strategies	This RAG system reconstructs the most valuable, contiguous text segments from the knowledge base by algorithmically identifying sequential chunks that collectively maximize relevance to the query.	Supplementay1 Fig. 11
Fusion RAG	Core Retrieval Strategies	This RAG system employs a hybrid retrieval workflow, retrieving candidate documents through parallel dense vector and sparse BM25 searches, and then re-ranks the combined results using	Supplementay1 Fig. 6

System	Category	System Description	Architecture
		a weighted fusion of their normalized scores to construct the final context.	
Context Compression RAG	Post-retrieval Refinement Strategies	This RAG system first retrieves a broad set of document chunks, then employs a language model to filter and condense these into a highly relevant context before final answer generation.	Supplementay1 Fig. 3
Rerank RAG	Post-retrieval Refinement Strategies	This RAG system initially fetches a broad set of text chunks through vector search and then has them re-ordered by a specialized reranker model to supply the most contextually relevant information to the language model.	Supplementay1 Fig. 10
Context Enrichment RAG	Post-retrieval Refinement Strategies	This RAG system enriches the retrieved context by augmenting each relevant chunk with its adjacent neighboring passages before answer generation.	Supplementay1 Fig. 4
Corrective RAG (CRAG)	Self-Corrective Frameworks	This RAG system assesses the relevance of initially retrieved documents for queries and dynamically augments the context with web search results when relevance is insufficient to ensure a well-grounded answer.	Supplementay1 Fig. 1
Self RAG	Self-Corrective Frameworks	This RAG system employs an iterative, self-correcting loop where the language model generates, critiques, and progressively refines its answer until quality criteria are met.	Supplementay1 Fig. 12

Evaluation Framework

Scoring Criteria

We evaluated answer quality using a four-level rubric adapted from our prior study[26]. Responses were categorized as A–D based on four domains: (1) answer relevance and coverage with respect to the clinical question; (2) evidence grounding and accuracy; (3) faithfulness to retrieved evidence and factual accuracy (no hallucinations); (4) synthesis and practical utility for the clinical questions. Detailed scoring criteria are listed in Table 2.

For analysis, we calculated the Overall Winrate for each system. This metric was derived from pairwise comparisons on each question, where a 'win' was awarded for a higher score and ties were disregarded. The Overall Winrate was then calculated as follows:

$$Overall\,Winrate = \frac{N_W}{N_W + N_L}$$

Where 'N_W' represents the cumulative number of wins for a given system against all other systems across all questions, and 'N_L' is the cumulative number of losses.

Expert Review Process

Two gynecological experts independently and anonymously reviewed responses of each system. Concordant grades were accepted; when grades differed, the reviewers discussed and issued a single consensus grade. This consensus-based ordinal scheme reduces subjectivity and supports robust comparisons across RAG systems.

Table 2
Scoring Process and Criteria.

Grade	Definition	Scores
A	Direct, complete, and well-supported: the answer fully addresses the question and covers all key aspects; every major claim is supported by correct, specific evidence; faithful to sources with no hallucinations or contradictions; clear, well-structured, and clinically useful.	3
B	Mostly correct and well-supported: the answer addresses the question with minor gaps; most major claims have correct evidence; little unsupported or slightly overstated statements that do not change the main conclusion; presentation is clear with minor omissions.	2
C	Incomplete or weakly supported: the answer misses important aspects of the question; several key claims lack evidence or misinterpret the sources; noticeable inaccuracies or internal inconsistencies; limited clinical utility.	1
D	Incorrect, ungrounded, or off-topic: the answer fails to address the question or is misleading; evidence is irrelevant or incorrect; major factual errors or hallucinations; unsafe recommendations.	0

Statistical Methods

Statistical analyses were performed in R version 4.5.1 (R Foundation for Statistical Computing, Vienna, Austria) using RStudio. Between-model differences in the distribution of categorical grades were assessed with Pearson's chi-square test of independence. For paired, within-question comparisons between two models, normality of paired differences was first assessed using the Shapiro–Wilk test; if normally distributed, a paired t test was applied; otherwise, a Wilcoxon signed-rank test was used. To compare the overall performance difference between the two question sets, a Wilcoxon signed-rank test was also applied to the paired mean scores of the 14 systems. The Overall Winrate was also calculated. All tests were two-sided, and $P < .05$ was considered statistically significant.

Declarations

Data availability

The datasets generated and analyzed during the current study is available at https://github.com/lilaotou2/Evaluated-RAG-Systems.

Code availability

The source code we use to perform RAG is available at https://github.com/lilaotou2/Evaluated-RAG-Systems.

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Contributions

Y.Y. conceptualized the work. Y.Y. and S.P. designed the study. Y.Y., S.P., and W.K. wrote codes and performed analyses. Y.Y., S.P., H.X., X.H., and W.K collected the data and evaluate the result. Y.Y., W.K. and S.P. wrote the first draft and revised the manuscript. All authors read and approved the final manuscript.

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Competing interests

All authors declare no financial or non-financial competing interests

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Figures

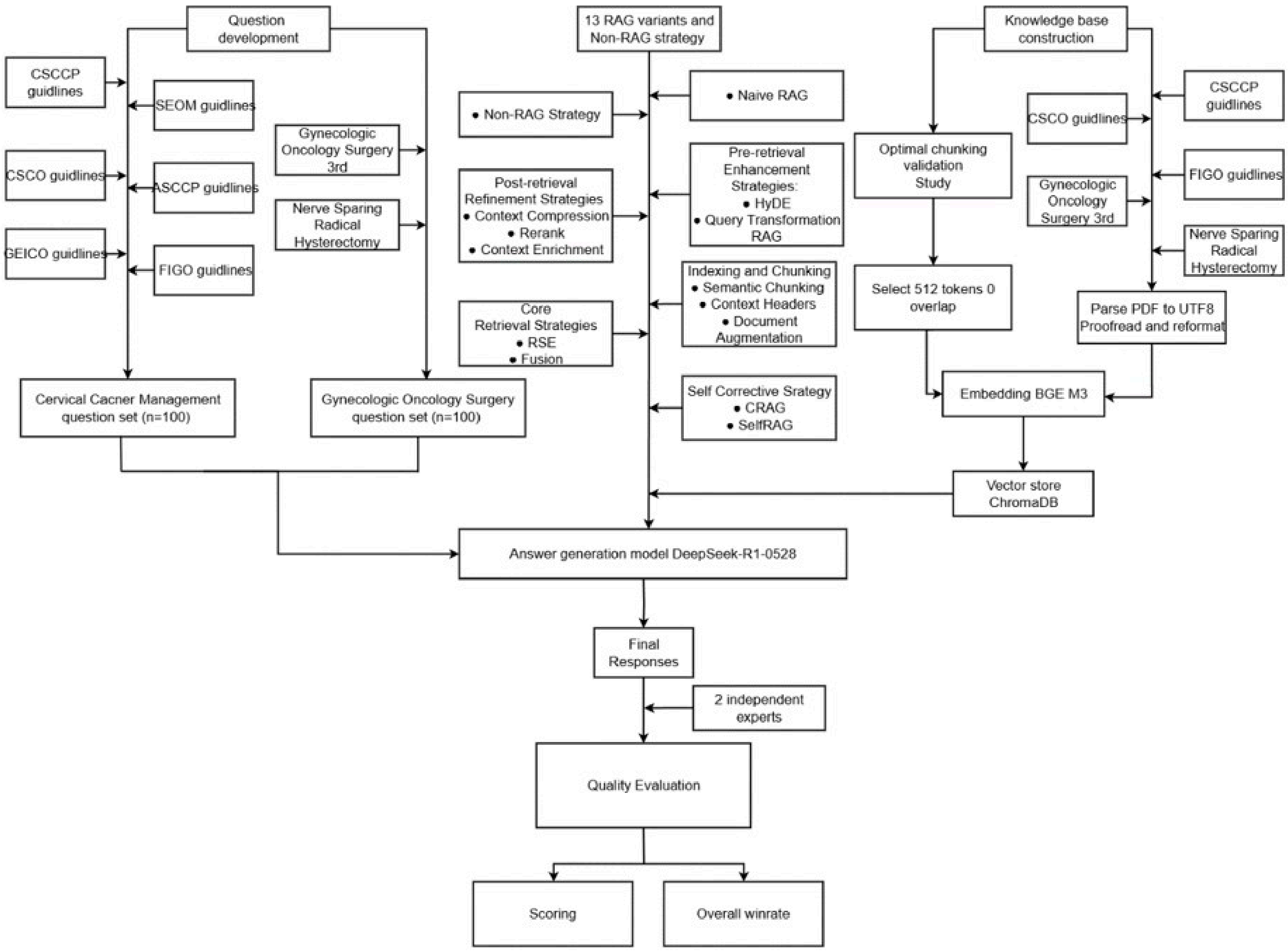

Figure 1

The standardized framework for the evaluation of Retrieval-Augmented Generation (RAG) systems in gynecologic oncology.

RAG Strategy	A	B	C	D	Mean (95%CI)	Win Rate
Semantic.Chunking	91	3	5	1	2.84 [2.73, 2.95]	67.67%
Hyde	90	5	1	4	2.81 [2.68, 2.94]	63.21%
SelfRag	87	10	0	3	2.81 [2.69, 2.93]	62.03%
Query.Transformation	88	4	6	2	2.78 [2.65, 2.91]	56.31%
Rerank	87	6	5	2	2.78 [2.66, 2.90]	55.60%
Fusion.Rag	88	4	6	2	2.78 [2.65, 2.91]	55.22%
Contextual.Compression	84	14	2	0	2.82 [2.73, 2.91]	54.55%
RSE	86	6	3	5	2.73 [2.58, 2.88]	51.23%
Contextual.Chunk.Headers	84	9	3	4	2.73 [2.59, 2.87]	47.22%
CRAG	82	12	6	0	2.76 [2.65, 2.87]	46.82%
Document.Augmentation	85	6	5	4	2.72 [2.57, 2.87]	46.64%
Naive.RAG	78	16	5	1	2.71 [2.59, 2.83]	40.12%
Context.Enrichment	81	7	7	5	2.64 [2.48, 2.80]	38.64%
Non-RAG.Strategy	74	16	7	3	2.61 [2.46, 2.76]	34.79%

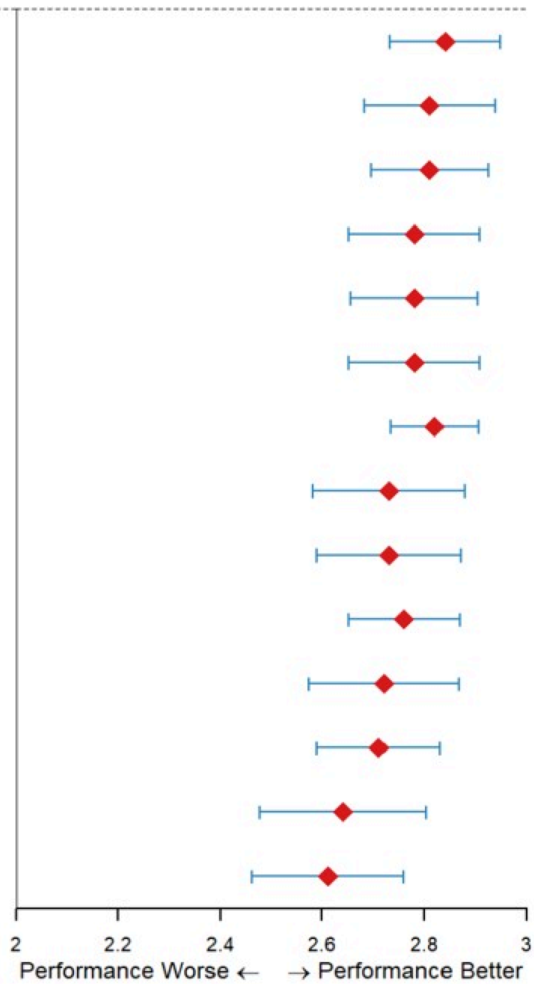

Figure 2

Performance assessment of various RAG systems on cervical cancer management questions.

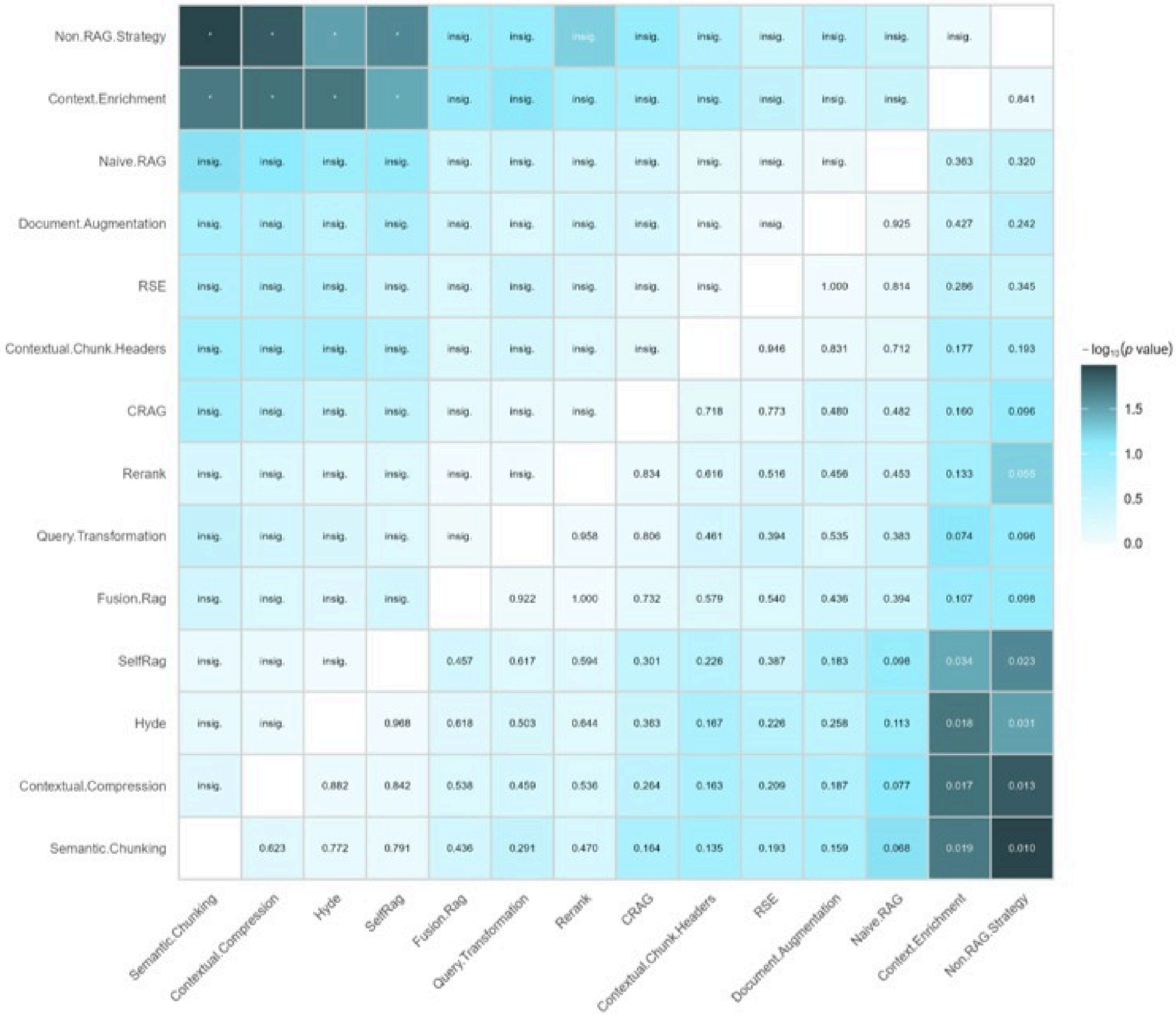

Figure 3

Heatmap of Pairwise Statistical Significance of Performance Across Different Methods on cervical cancer management question set.

RAG Strategy	A	B	C	D	Mean (95%CI)	Win Rate
Contextual.Compression	82	7	6	5	2.66 [2.50, 2.82]	84.02%
RSE	71	12	4	13	2.41 [2.20, 2.62]	68.74%
SelfRag	64	22	8	6	2.44 [2.27, 2.61]	66.67%
CRAG	62	22	14	2	2.44 [2.28, 2.60]	66.17%
Fusion.Rag	68	13	6	13	2.36 [2.15, 2.57]	63.48%
Document.Augmentation	62	14	9	15	2.23 [2.01, 2.45]	57.59%
Contextual.Chunk.Headers	60	16	9	15	2.21 [1.99, 2.43]	54.41%
Rerank	57	15	8	20	2.09 [1.85, 2.33]	49.40%
Query.Transformation	50	23	13	14	2.09 [1.87, 2.31]	43.24%
Hyde	50	22	14	14	2.08 [1.86, 2.30]	42.88%
Semantic.Chunking	47	26	14	13	2.07 [1.86, 2.28]	39.17%
Naive.RAG	51	20	10	19	2.03 [1.80, 2.26]	37.43%
Context.Enrichment	50	13	7	30	1.83 [1.57, 2.09]	31.61%
Non-RAG.strategy	18	32	31	19	1.49 [1.29, 1.69]	20.78%

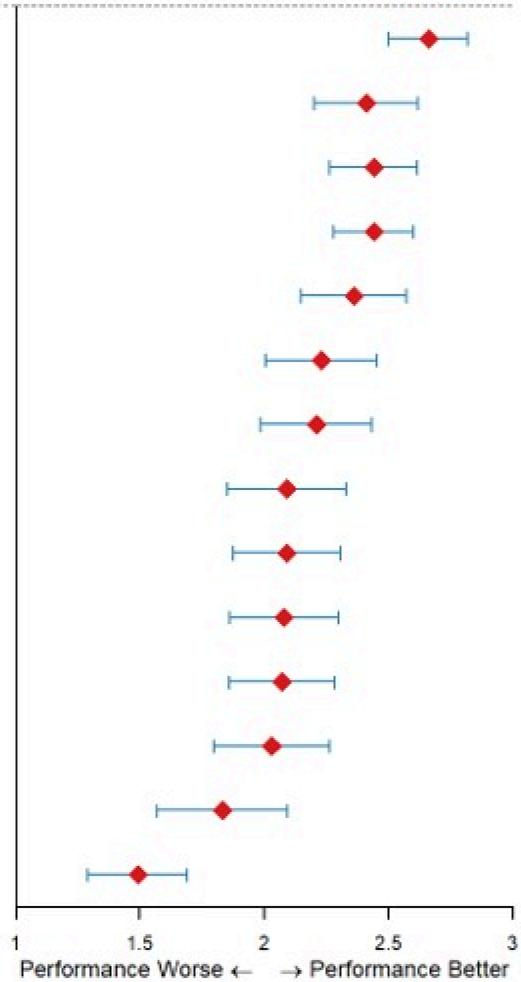

Figure 4

Performance assessment of various RAG systems on gynecological surgery questions.

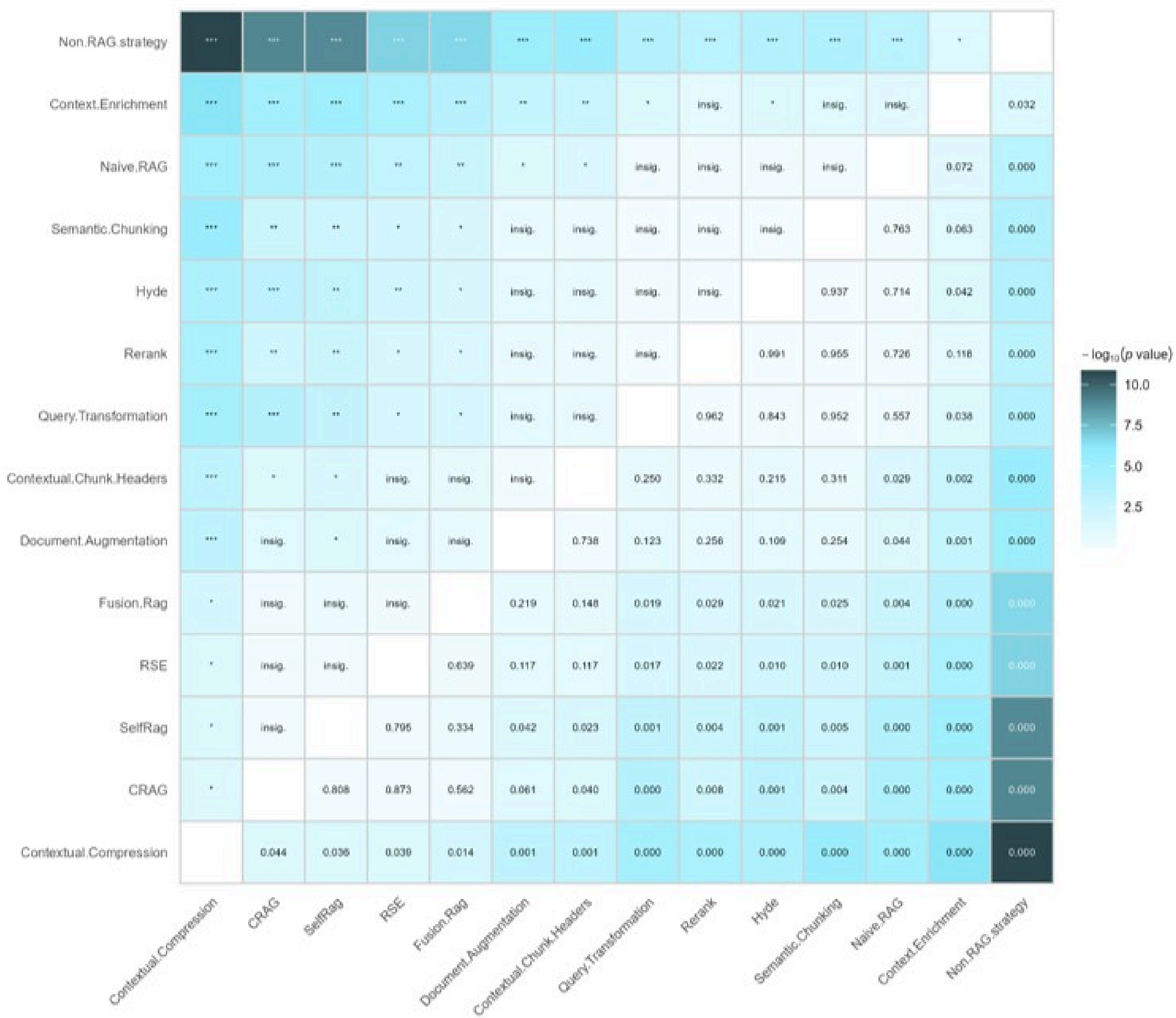

Figure 5

Heatmap of Pairwise Statistical Significance of Performance Across Different Methods on gynecological surgery question set.

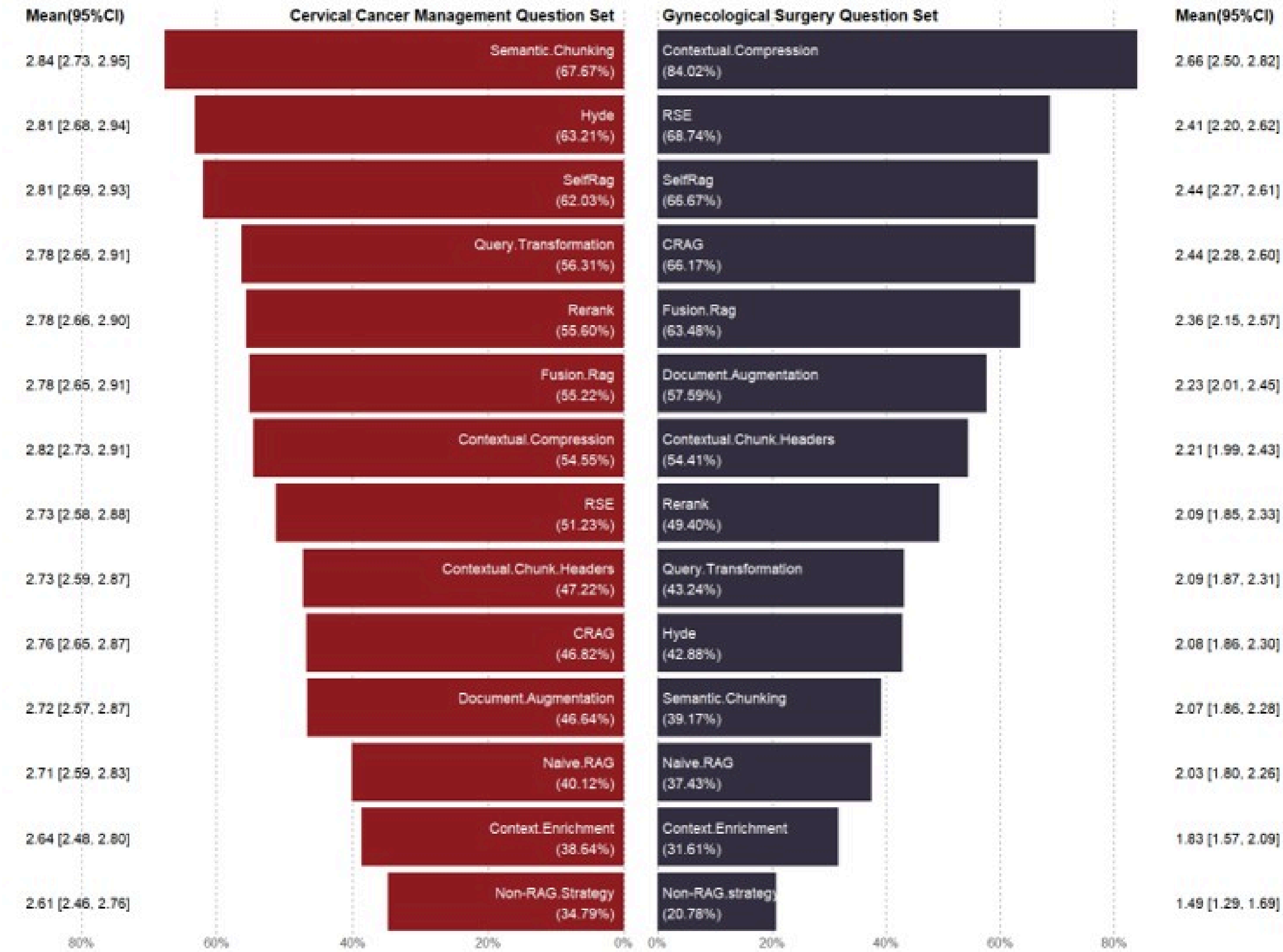

Figure 6

Comparative performance of RAG strategies across two medical question sets.

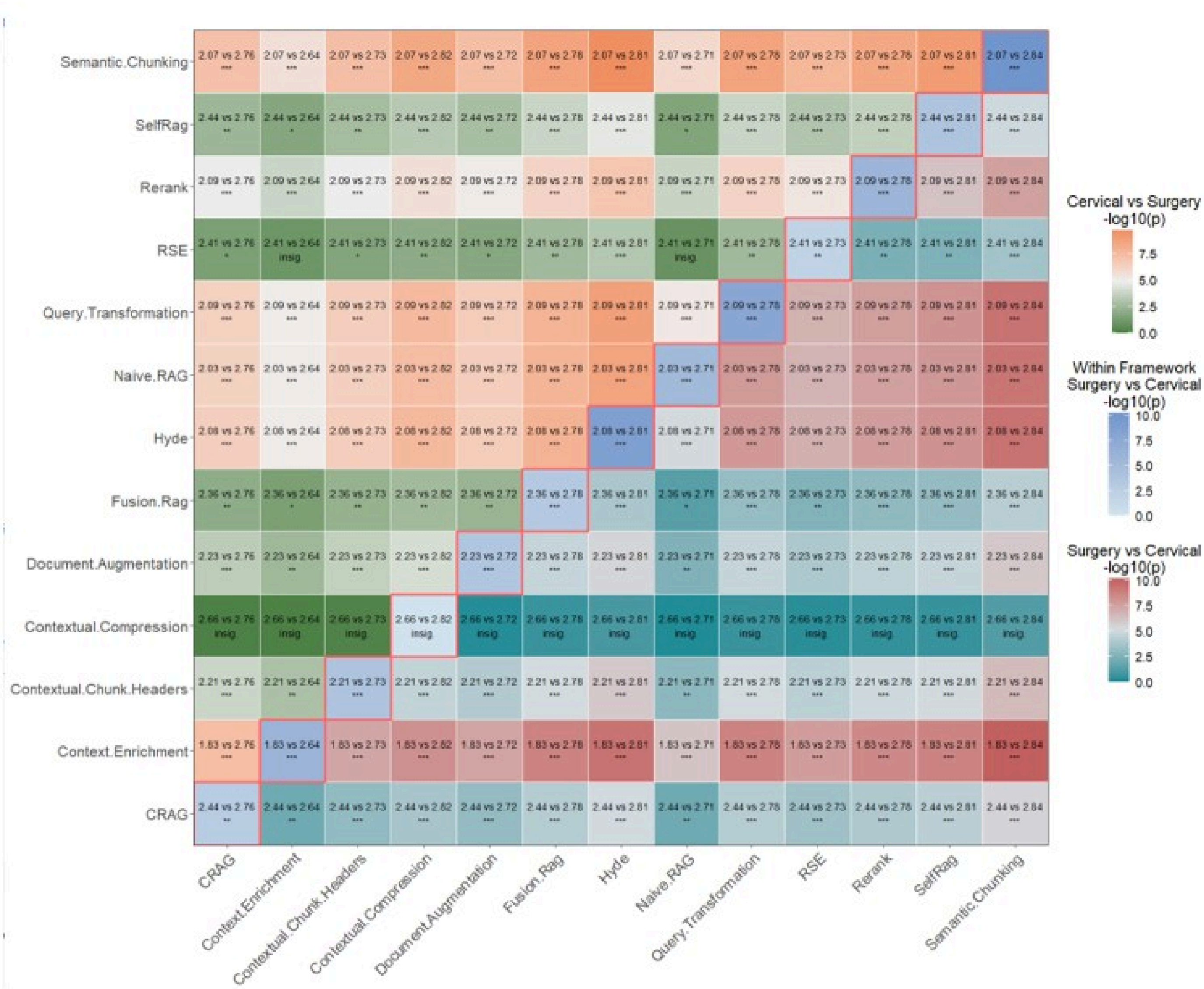

Figure 7

Heatmap of cross-domain significance (Mann–Whitney U).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Supplementary1.docx
- Supplementary2.docx
- Supplementary3.docx